

JEE MAIN REVISION MASTER: CENTER OF MASS AND COLLISION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 128 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Center of Mass Calculation & Cut-out Sections	74 PYQs	Center of Mass
Very High	Conservation of Linear Momentum & Explosion	56 PYQs	Momentum

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Center of Mass Calculation & Cut-out Sections	$\vec{r}_{\text{cm}} = \frac{\sum m_i \vec{r}_i}{\sum m_i}, \quad \vec{r}_{\text{remaining}} = \frac{M\vec{r}_0 - m\vec{r}_{\text{removed}}}{M - m}$
Conservation of Linear Momentum & Explosion	$\vec{F}_{\text{ext}} = M\vec{a}_{\text{cm}}, \quad \text{If } \vec{F}_{\text{ext}} = 0 \implies \vec{v}_{\text{cm}} = \text{const}, \quad \sum m_i \vec{v}_i = \text{const}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Center of Mass Calculation & Cut-out Sections (Very High Yield — 74 PYQs)

Core Principle: Discrete and continuous center of mass coordinates, and negative mass method for cavity problems.

■ **Caution / JEE Pitfall:** Hemisphere CM is at $3R/8$ from center, whereas hemispherical shell CM is at $R/2$.

■ **Speed Trick:** Treat a removed cavity as negative mass at the cavity's geometric center: $X_{\text{(new)}} = (A_1 x_1 - A_2 x_2) / (A_1 - A_2)$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Two identical bodies A and B of equal masses have initial velocities $\vec{v}_1=4\hat{i}\text{ m/s}$ and $\vec{v}_2=4\hat{j}\text{ m/s}$ respectively. T..." **[Ans: C]**
- **[2026 Morning Shift] Q#2 (MCQ):** "The position of center of mass of three masses 2kg, 3kg and 15 kg placed with respect to mid point (p) of normal bisector, as shown in th..." **[Ans: A]**
- **[2026 Morning Shift] Q#2 (MCQ):** "A small bob A of mass m is attached to a massless rigid rod of length 1 m pivoted at point P and kept at an angle of 60° with..." **[Ans: A]**

3.2 Conservation of Linear Momentum & Explosion (Very High Yield — 56 PYQs)

Core Principle: Motion of center of mass under zero net external force and explosive fragmentation trajectories.

■ **Caution / JEE Pitfall:** Internal forces (like explosion or spring expansion) cannot change the trajectory of the center of mass.

■ **Speed Trick:** If a projectile explodes mid-air, the center of mass continues along the original parabolic trajectory until fragments hit the ground.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "In a perfectly inelastic collision, two spheres made of the same material with masses 15 kg and 25 kg, moving in opposite directions with s..." **[Ans: B]**
- **[2026 Evening Shift] Q#4 (MCQ):** "A body of mass 14 kg initially at rest explodes and breaks into three fragments of masses in the ratio 2:2:3. The two pieces of equal mas..." **[Ans: N/A]**
- **[2025 Evening Shift] Q#1 (MCQ):** "Consider two blocks A and B of masses $m_1=10\text{kg}$ and $m_2=5\text{kg}$ that are placed on a frictionless table. The block A moves with a constan..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Evening Shift (Q#1)

Two identical bodies A and B of equal masses have initial velocities $\vec{v}_1=4\hat{i}\text{ m/s}$ and $\vec{v}_2=4\hat{j}\text{ m/s}$ respectively. The body A has acceleration $\vec{a}_1=6\hat{i}+6\hat{j}\text{ m/s}^2$ while the acceleration of the other body B is zero. The centre of mass of the two bodies moves in _____ path.

- (A) circular
- (B) parabolic
- (C) straight line **[Correct]**
- (D) elliptical

Correct Answer: C

Solution Analysis: For two bodies of equal mass ($m_1=m_2=m$), the velocity of the COM is:

$\vec{v}_{cm} = \frac{m\vec{v}_1 + m\vec{v}_2}{m+m} = \frac{\vec{v}_1 + \vec{v}_2}{2}$ The velocity of body A is $\vec{v}_1=4\hat{i}$ and that of body B it is $\vec{v}_2=4\hat{j}$. So, $\vec{v}_{cm} = \frac{4\hat{i} + 4\hat{j}}{2} = (2\hat{i} + 2\hat{j})\text{ m/s}$ Similarly, for equal masses the acceleration of center of mass is, $\vec{a}_{cm} = \frac{\vec{a}_1 + \vec{a}_2}{2}$ The acceleration of body A is $\vec{a}_1=6\hat{i}+6\hat{j}$ and that of body B is $\vec{a}_2=0$: $\vec{a}_{cm} = \frac{(6\hat{i}+6\hat{j})+0}{2}$

Benchmark Problem 2: JEE Main 2026 (Online) 23rd January Morning Shift (Q#1)

In a perfectly inelastic collision, two spheres made of the same material with masses 15 kg and 25 kg, moving in opposite directions with speeds of 10m/s and 30m/s, respectively, strike each other and stick together. The rise in temperature (in °C), if all the heat produced during the collision is retained by these spheres, is : (specific heat of sphere material 31cal/kg.°C and 1cal=4.2J)

- (A) 1.75
- (B) 1.44 [Correct]
- (C) 1.95
- (D) 1.15

Correct Answer: B

Solution Analysis: Before Collision : The masses of the spheres are, $m_A=15\text{kg}$, $m_B=25\text{kg}$ The initial velocities of the spheres are, $u_A=10\text{m/s}$, $u_B=-30\text{m/s}$ Hence, the kinetic energy of system before collision is, $K_i = \frac{1}{2}m_A u_A^2 + \frac{1}{2}m_B u_B^2 \Rightarrow K_i = \frac{1}{2}(15)(10)^2 + \frac{1}{2}(25)(30)^2 = 750 + 11250 = 12000\text{J}$ Since the collision is perfectly inelastic, the spheres stick together and move with a common final velocity v_f . Using conservation of linear momentum, $m_A u_A + m_B u_B = (m_A + m_B)v_f \Rightarrow (15)(10) + (25)(-30) = (15+25)v_f \Rightarrow v_f = -15\text{m/s}$